

EE 3201.302

Lab 5 PSpice, Simulation of Circuits

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I. Abstract

In this lab, we will utilize PSpice to simulate and analyze a wide range of circuit configurations, covering both simple and complex designs. The primary objective is to gain proficiency with the design, simulation, and analysis tools available within the PSpice environment, including the Bias Point, DC Sweep, AC Sweep, and Parametric Sweep functions. By conducting these simulations, we aim to deepen our understanding of core circuit simulation concepts, such as voltage, current, and impedance behavior across different conditions. The simulation results will be compared with hand calculations to verify accuracy and provide a more comprehensive understanding of the practical behavior of circuits, helping to develop skills in both theoretical and applied circuit analysis.

II. Introduction

The pre-laboratory exercises guide the user through key simulations, starting with a Bias Point analysis to determine the operating point of simple DC circuits. DC Sweep simulations explore the circuit's large-signal behavior by observing its response to input voltage changes, while AC Sweep simulations provide insights into its small-signal behavior and frequency response. The exercises conclude with a Parametric Sweep, enabling users to examine how changes in circuit parameters, such as resistor values or supply voltages, impact performance. These simulations highlight the importance of understanding how factors influence circuit behavior and emphasize the role of simulation in design. By using PSpice to simulate real-world conditions, users gain experience in designing and optimizing circuits effectively.

III. Procedure

1. Boost Converter
 - a. Build Circuit

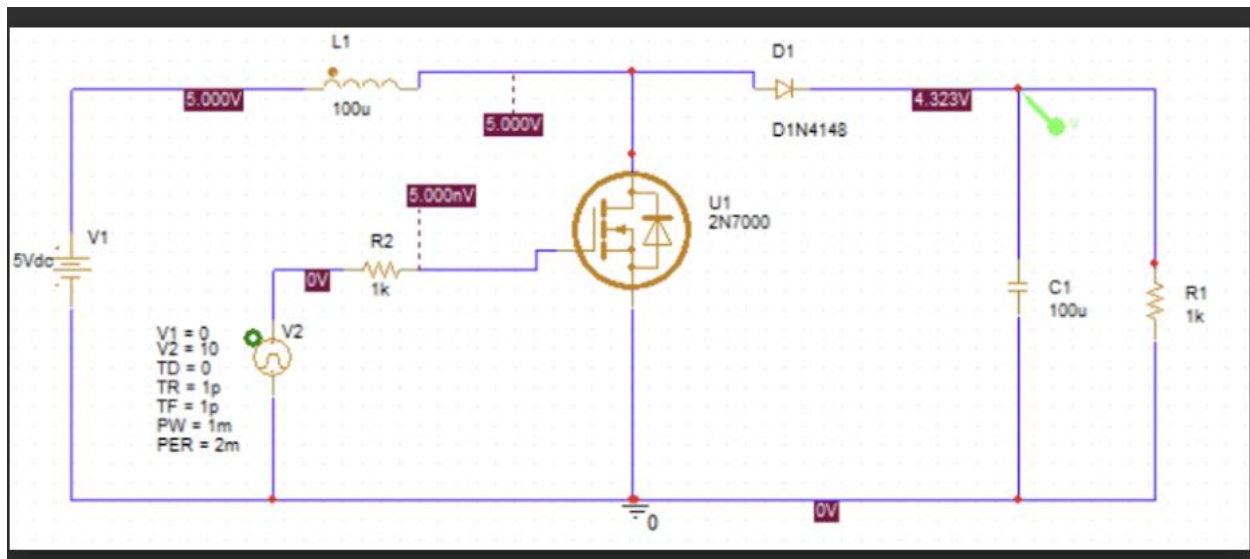


Figure 1.1: Boost Converter Circuit

b. Simulation

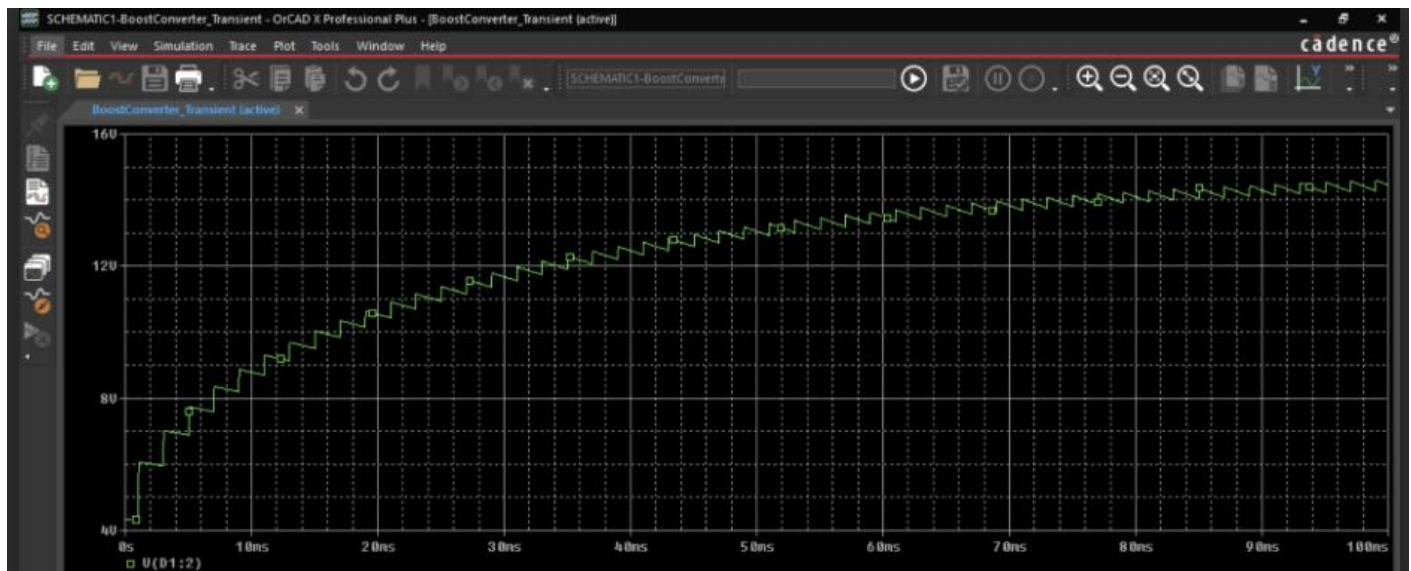


Figure 1.2: Boost Converter Simulation

2. Buck Converter

a. Build Circuit

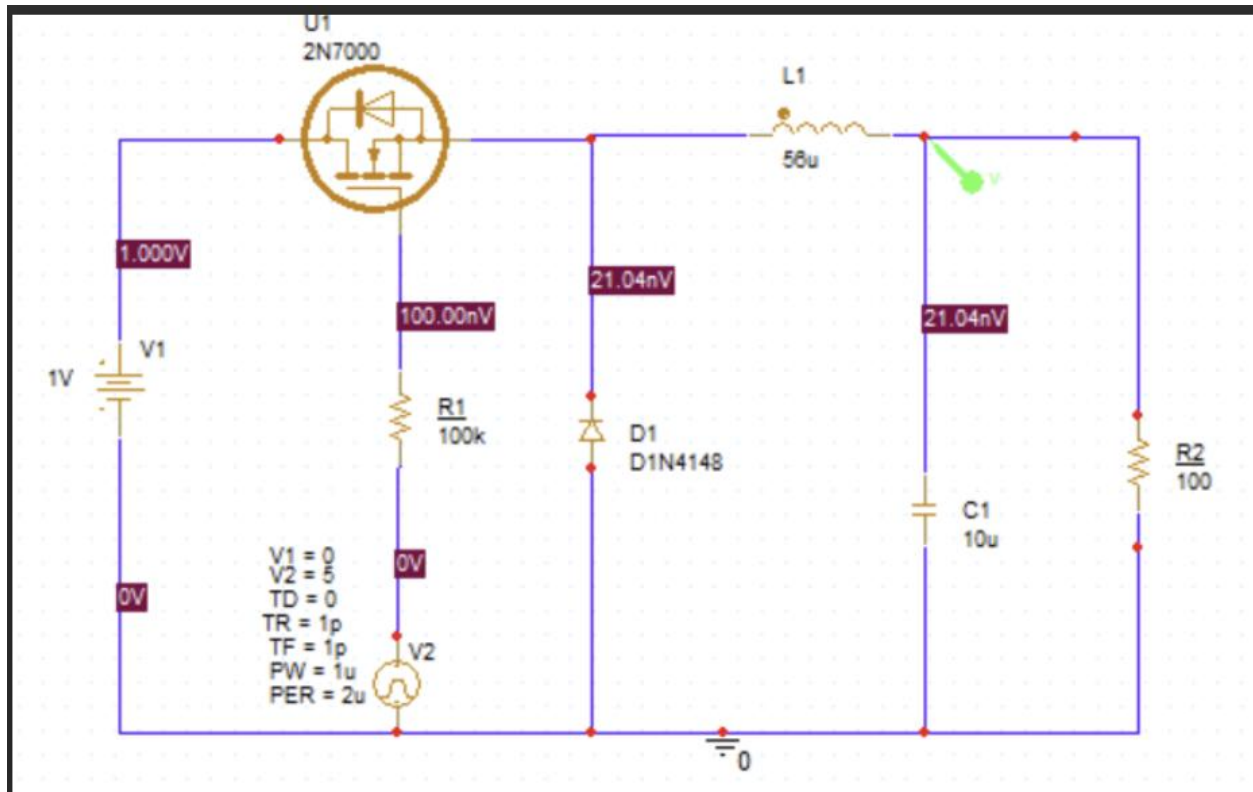


Figure 2.1: Buck Converter Circuit

b. Simulation

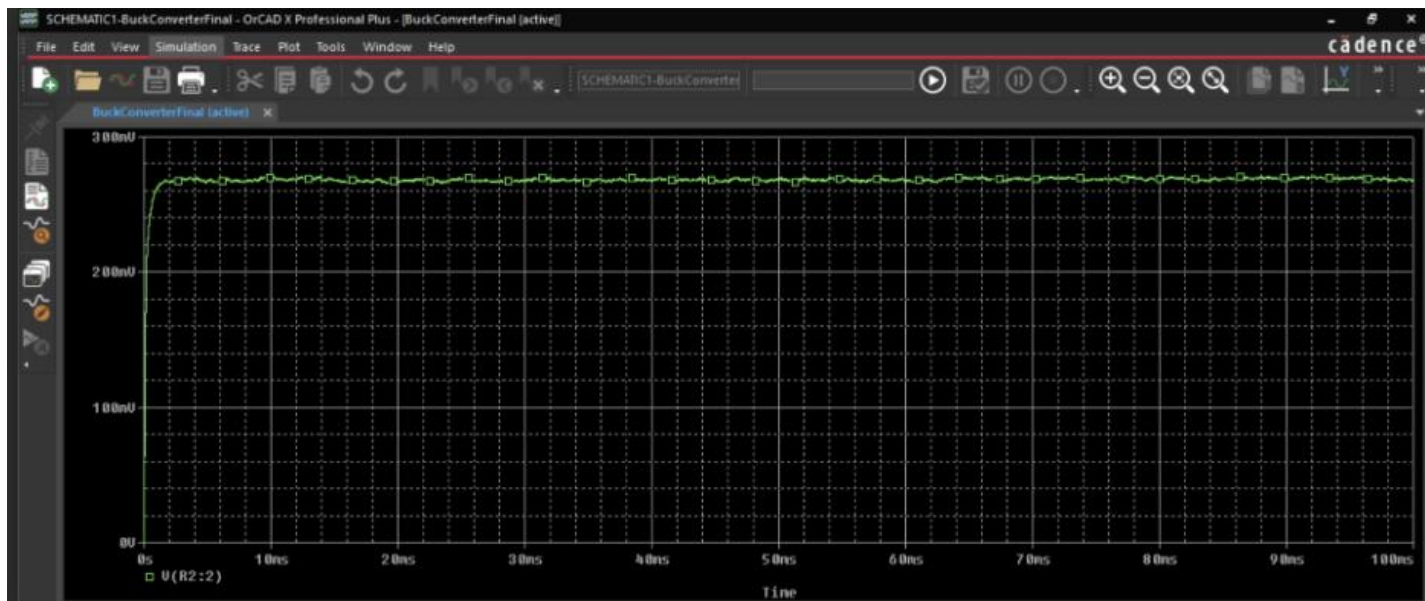


Figure 2.2: Buck Converter Simulation

c. Physical Circuit

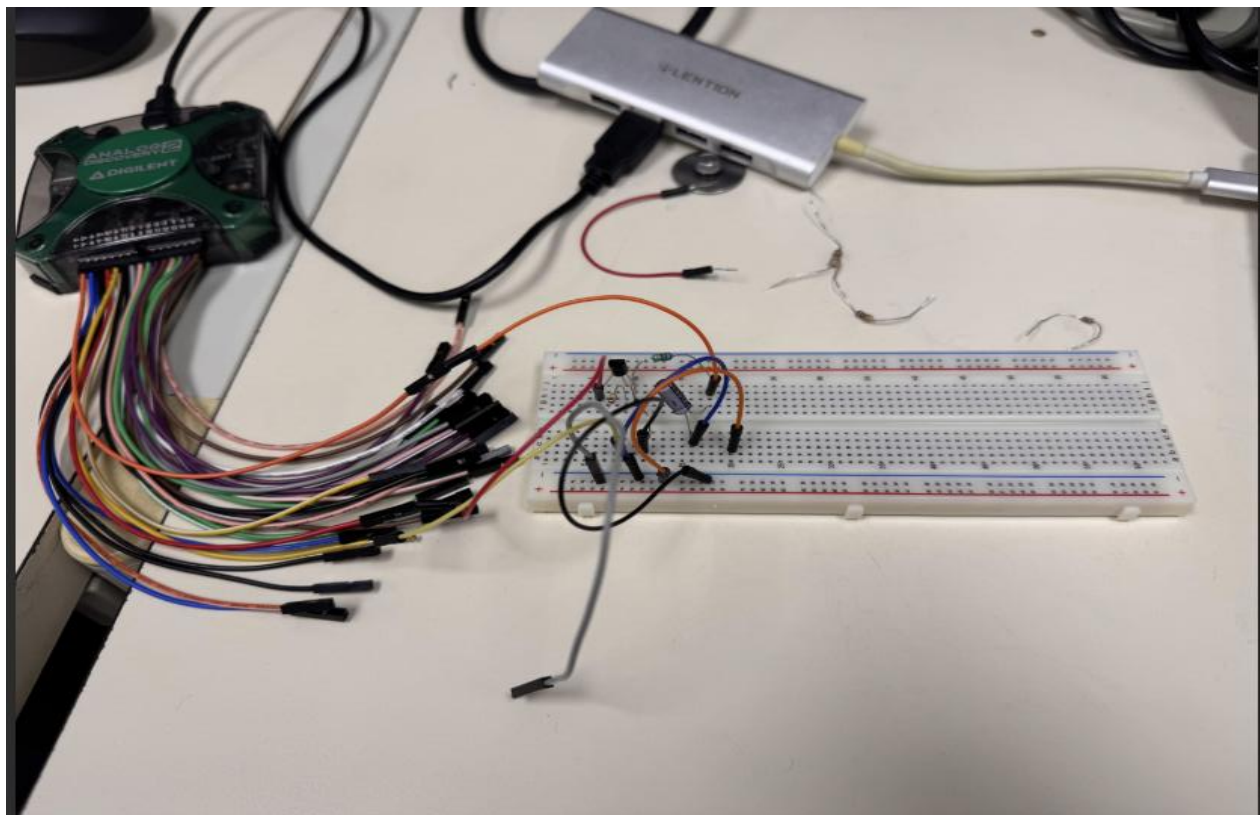


Figure 2.3: Buck Converter Physical Circuit

d. Physical Circuit Simulation

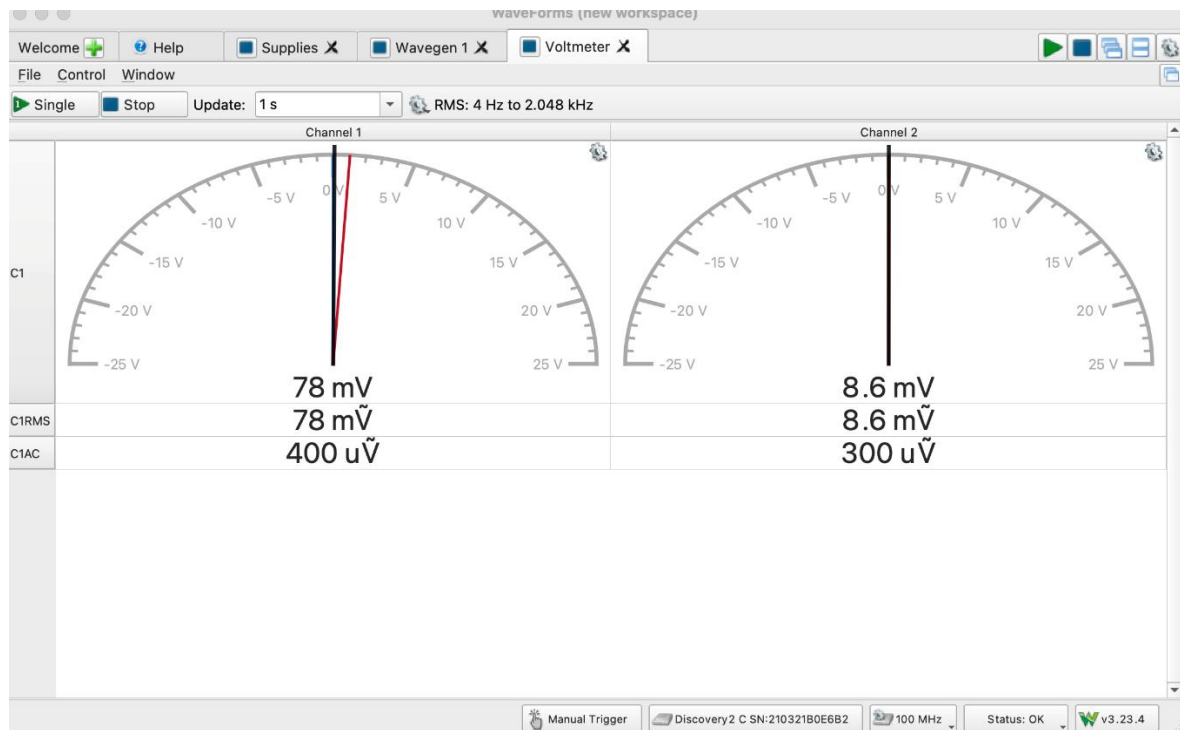


Figure 2.4: Buck Converter with 10 ohms

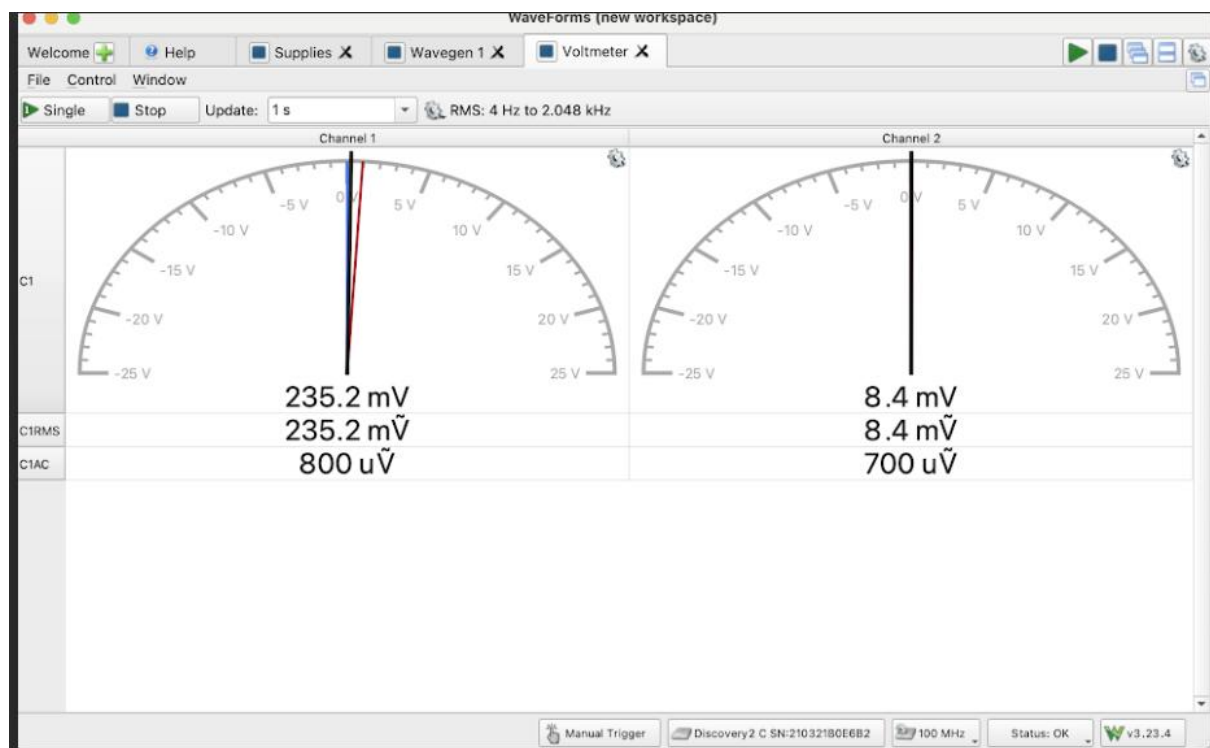


Figure 2.5: Buck Converter with 100 ohms

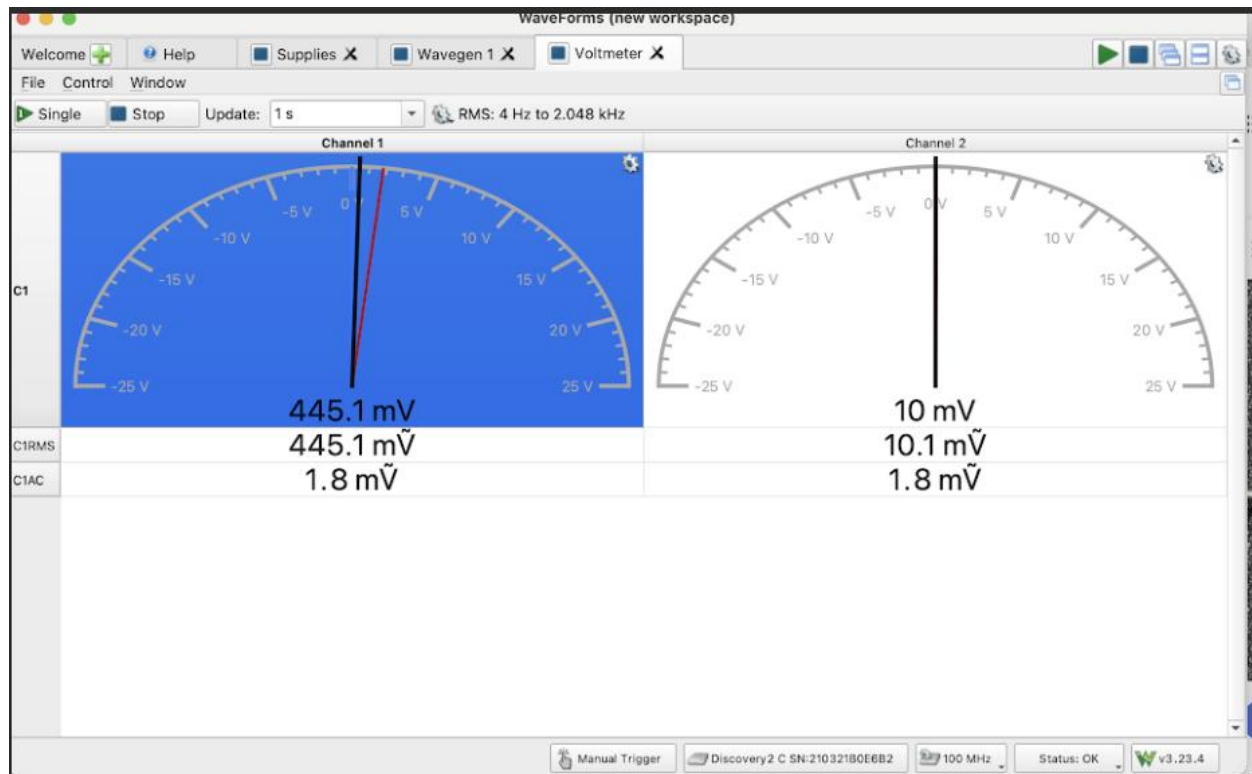


Figure 2.6: Buck Converter with 1000 ohms

e. Observations

Resistor values (ohm)	Simulation Output voltage (mV)	Physical Circuit output voltage (mV)
10	95	78
100	255	235
1000	460	445

Table. 2.7: Comparison between simulation output voltage and physical output voltage.

The physical buck converter circuit output voltage is lower than the simulation output voltage. As we saw, for each resistance measurement, we can justify it is due to the tolerance of the lab equipment and components used. Another observation is that the results are more accurate as the resistance increased.

IV. Experimental Results

1. Boost Converter

The Boost converter circuit was both built and simulated in PSpice, where the results confirmed its voltage conversion characteristics. With a fixed input voltage, the output voltage was monitored across various simulation conditions. As expected, the boost converter successfully stepped up the input voltage. The output voltage increased in line with the chosen component values and duty cycle, confirming proper operation of the converter. The observed output voltage exceeded the input voltage, consistent with the expected behavior of a boost converter. Using the voltage conversion equation, which factors in input voltage, duty cycle, and inductance, the theoretical output voltage was calculated. The simulation results closely matched these theoretical values, validating the accuracy of the PSpice model in representing the circuit's behavior.

2. Buck Converter

The Buck converter was analyzed through both PSpice simulation and physical implementation. The primary objective was to observe how varying load resistance impacts the output voltage. As a step-down DC-DC converter, the buck converter's behavior was simulated to verify its expected performance under different load conditions. In the PSpice simulation, the input voltage was kept constant, and steady-state conditions were applied to ensure accurate results. The simulation produced different output voltages corresponding to each resistor value. These results aligned with theoretical expectations, showing that as the load resistance increased, the output voltage also rose. This behavior is typical of buck converters, where a higher load resistance results in reduced current flow, which in turn leads to a higher output voltage due to the converter's design.

V. Post-lab Questions

1. The dampening factor is a value which represents the sensitivity of a circuit to return to equilibrium. If a circuit was oscillating the dampening factor would represent how quickly the circuit returned to equilibrium.
2. A boost converter is a dc-dc converter which will boost voltage and reduce the current all while keeping the same power. The boost converter is an effective power conversion circuit.
3. Boost converters work by storing energy in an inductor's magnetic field. Once the switch is opened the stored voltage is added to the input voltage. A diode is also included to prevent backflow current. If we adjust the duty cycle of the switch, we can control the output voltage.
4. Boost converters have several applications, but the most common uses are LED drivers, power supplies for portable electronics like laptops.
5. A buck converter is a dc-dc converter which will reduce the voltage and increase the current all while keeping the same power. It is a power conversion circuit.

6. A buck converter works with a switch and inductor as well, but in this case the inductor is releasing the energy when the switch is open. The current flow through the diode is maintained. A lower duty cycle means lower output voltage.

7. Buck converters are used in many applications such as automotive electronics, battery chargers, and low-voltage logic circuits.

8. The average percentage error for all three resistors is around 11.2%. For the 10 ohm is 21.7%. The 100 ohm is 8.5%. 1000 ohm is 3.4%. I noticed the percentage error decreased as the resistance increased.

9. In my opinion the most difficult part of the lab was the learning curve to get used to Pspice after figuring out the boost converter the buck converter worked with ease.

VI. Discussion and Conclusion

In this lab, we investigated the design, simulation, of both a Boost and a Buck converter circuit. The primary goal was to compare the behavior of these circuits in simulation using PSpice with their real-world counterparts. By performing simulations and then building and testing physical circuits for the buck converter, we were able to analyze how each converter operates and how closely simulation results match physical measurements.

For the boost converter, the simulation accurately predicted the expected output voltage based on the chosen duty cycle and load conditions. The converter successfully increased the input voltage to a higher output voltage, as anticipated. At first, we ran into an issue with an incorrect graph, but we later came to know that our simulation settings were incorrect.

In the Buck converter simulation, the output voltage consistently followed the expected trend, increasing as the load resistance increased. When comparing the simulation with the physical circuit results, we observed that the physical measurements were slightly lower than the simulated values. This difference was likely due to real-world factors such as component tolerances and measurement error, but the general trends in both simulation and physical circuit were aligned.

Through the PSpice simulation, we gained deeper insight into how the duty cycle, inductance, and capacitance affect the performance of DC-DC converters. Learning how to use PSpice has given us a tool to simulate any circuit we want for future use. In conclusion, the lab provided comprehensive learning experience in both simulation and physical implementation of DC-DC converters. It successfully demonstrated the utility of PSpice for circuit analysis and gave us experience in troubleshooting, adjusting, and verifying circuit designs in the laboratory setting.